

Documentation for finddeltaminimaxdesign.cpp

This file contains all the functions necessary to build a binary file. I have only tested it on a SPARC computer, so it may well need modifications to run in different environments.

Once built, the binary file takes seven arguments:

1. δ_0 - a real number representing the mean effect of the control treatment (and the new treatment under the null hypothesis).
2. δ_1 - a real number representing the clinically relevant difference, i.e. the mean effect of the new treatment under the alternative hypothesis. The value of δ_1 should be higher than δ_0 .
3. σ - a positive real number representing the standard deviation of the treatment effect in both arms.
4. a positive real number between 0 and 1 representing the required type I error of the design.
5. a positive real number between the type I error and 1 representing the required power of the design.
6. an integer representing the number of stages of the design (only tested for integer values between 2 and 5).
7. a string representing the name of the file for which the δ -minimax design should be written to.

These arguments must be given in the right order and in the right format. For example:

```
./finddeltaminimaxdesign 0 1 2 0.05 0.9 2 temp
```

will search for the two-stage δ -minimax design with type I error 0.05, power 0.9, $\delta_0 = 0$, $\delta_1 = 1$, and $\sigma = 2$.

The program will print out the current maximum expected sample size after each restart of the process. Two separate processes are ran, the first one allowing the group-size to be a non-integer, and the second one fixing the group-size at the floor of the optimal group-size from the first process.

The simulated annealing parameters provided are the default ones used in the paper. They can be modified, and if you are unsure what the different ones do, then please get in contact with me. If there is sufficient demand, I may modify the code so it is easier to modify the simulated annealing parameters, and provide documentation of what each one does.